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Mohammad Dastgheib

PhD Candidate in Psychology and Cognitive Neuroscience

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I am a Cognitive Neuroscience PhD Candidate who quantifies human perception and decision-making. I specialize in computational modeling (DDM, psychometric functions) and physiological measures (pupillometry) to objectively assess cognitive load and user behavior. With a proven ability to lead collaborative, multi-institutional research, I translate complex data into actionable insights for both academic publications and real-world applications.

Education

2021 – 2026 **PhD Candidate, Psychology**, *University of California-Riverside*, Riverside, CA
Preliminary thesis: 'A Mechanistic Investigation of Physical-Cognitive Effort Interactions and Their Modulation by Aging'
Supervisors: Ilana J. Bennett, PhD (UC Riverside); Aaron R. Seitz, PhD (Northeastern University)
GPA: 3.93

2017 – 2020 **M.Sc., Psychology**, *Queen's University*, Kingston, ON
Thesis: 'The Effects of Self-Guided Meditation and Napping on Memory Consolidation in Humans'
Supervisor: Hans C. Dringenberg, PhD
GPA: 3.83

2013 – 2017 **Hon. B.Sc., Biological Sciences & Psychology**, *York University*, Toronto, ON, Double major
Thesis: 'Are Older Adults' Attitudes about Aging Influenced by a Psychoeducational Program?'
Supervisors: Jill B. Rich, PhD, CPsych (York University); Angela K. Troyer, PhD, CPsych (Baycrest Hospital)

2008 – 2012 **B.Eng., Chemical Engineering**, *Azad University*, Tehran, IRN, *Transferred*

Skills

UX & XR Research	User interviews & moderation; usability testing; survey design (Qualtrics); mixed-methods; research ops; Extended Reality (XR) experimental design; adaptive modality frameworks
Data & Analytics	Computational Modeling (Drift Diffusion Models/DDM, Hierarchical Bayesian); Time-series features; gradient-boosted models; anomaly detection; experiment design (A/B); explainability (SHAP)
Behavior & Biosignals	Pupillometry, HRV, blink rate, grip/tremor; attention & fatigue modeling; signal simulation for prototyping
Programming	R (Shiny, tidyverse, ggplot2, plotly); Python (pandas, PyTorch); SQL; Bash; AI-Assisted Development (Cursor, LLM workflows)
Tooling & Deployment	Docker; CI/CD (GitHub Actions); ShinyApps.io; renv; testthat; Quarto; Git/GitHub; LaTeX; Rapid Prototyping
Languages	English (Proficient), Persian (Native), French (Basic)

Research & Professional Experience

2023–Present **Graduate Student Researcher**, *Soundmind collaboratary*, Northeastern University
- Validated neuropsychological tests
- Trained and supervised three research assistants

2022–Present **Graduate Student Researcher**, *LANI lab*, UC Riverside

- Analyzed Diffusion MRI data using FSL, including Probabilistic Tractography and Tensor-Based Metrics
- Trained and supervised three research assistants

2021–Present **Graduate Student Researcher**, *Seitz lab*, UC Riverside

- Designed and optimized perceptual and memory experiments using PsychToolbox
- Conducted extensive data analysis using R for behavioral and psychophysical data
- Preprocessed and analyzed pupillometry data using MATLAB, R, and Python
- Coordinated research projects and supervised two research assistants
- Collected data from over 80 participants

2021 **Research Assistant I**, *Surgical Safety Technologies (SST)*, Toronto, ON

Artificial Intelligence Annotator

- Assisted in the development of a state-of-the-art instance segmentation module for surgical tool detection and tracking
- Applied algorithms for quantifying inter-rater reliability metrics amongst the participating surgical analysts
- Conducted an in-depth surgical review of relevant healthcare papers in the machine learning domain

2017 – 2019 **Graduate Research Assistant**, *Neuroplasticity lab*, Queen's University

- Conducted data collection and analysis for over 60 participants
- Analyzed behavioral data using R and Bayesian methods
- Assessed sleep stages through EEG/EMG/EOG recordings
- Supervised volunteers and two research assistants
- Managed polysomnography equipment and protocols

2015 – 2020 **Research Fellow**, *Toronto General Hospital*, Toronto, ON

- Conducted data collection and analysis in the Internal Medicine Department, digitizing paper documents and managing patient records.
- Utilized Electronic Medical Records (EMR) system for data retrieval and analysis, ensuring data integrity and confidentiality.
- Contributed to clinical research through data management, analysis, and literature reviews.

2017 **Research Volunteer**, *Baycrest Hospital*, North York, ON

- Assisted organizers in educational seminars for the Memory & Aging program, contributing to knowledge dissemination in geriatric care.
- Analyzed older adults' behavioral performance post-workshop, aiding cognitive health assessments.
- Supported research initiatives on cognitive health in older adults, aiding evidence-based interventions.

2016 **Keenan Research Summer Student**, *St. Michael's Hospital*, Toronto, ON

- Shadowed administration of neuropsychological tests to older adults with MCI, including MoCA and MMSE.
- Participated in case discussions with Neurologists and Neuropsychologists.

2008 – 2009 **Undergraduate Research Volunteer**, *Tarbiat Modares University*, Tehran, IRN

- Conducted literature review in the department of chemical engineering.

Publications

Review paper

2022 **Dastgheib, M.**, *Kulanayagam, A., & Dringenberg, H. C.*, Is the role of sleep in memory consolidation overrated?, *Neuroscience & Biobehavioral Reviews, 140*, 104799
https://doi.org/10.1016/j.neubiorev.2022.104799

Preprints

2026 **Dastgheib, M.**, *& Pourmahdian, F.*, The Midas Touch in Gaze vs. Hand Pointing: Modality-Specific Failure Modes and Implications for XR Interfaces, *arXiv:2603.15991 [cs.HC]*
https://arxiv.org/abs/2603.15991

Under review

2026 **Dastgheib, M.,** Sun, A., Yaghoubi, K., Peters, M., Zhang, W., Bennett, I., Seitz, A., *Effects of Concurrent Task Difficulty and Physical Effort on Memory and Perceptual Performance and Pupil Response in Younger Adults*

In preparation

2026 **Dastgheib, M.,** Bennett, I., Seitz, A., *Physical Effort Selectively Impairs Evidence Accumulation Without Altering Response Caution in Older Adults: A Diffusion Model Analysis.*

2026 Sun, A.**, Dastgheib, M.,** Zhang, W., Maniscalco, B., Peters, M., Seitz, A., *Modulation of Perception and Metacognition by Hand Grip-Based Arousal in Older Adults.*

Media Coverage

2024 **The Boston Globe**, *Article: Five facts about sleep everyone should know*

- Work cited in mainstream media coverage of sleep research
- Published: September 23, 2024
- Read article

2023 **The Harvard Gazette**, *Article: Researchers one step closer to understanding daydreams*

- Research findings featured in Harvard University's official news publication
- Published: December 2023
- Read article

2024 **Curiosity Daily Podcast (Discovery)**, *Episode: Daydream Brain, Power of a Laugh, Caffeine & Soccer*

- Featured and discussed findings from my review article: *Is the role of sleep in memory consolidation overrated? Neuroscience & Biobehavioral Reviews* (2022).
- Original release: Jan 25, 2024; Re-release: Dec 19, 2024.
- Listen here

Selected Software & Research Projects

2024–Present **Pupil-DDM Framework**, *GitHub Repo*, Python, R, Hierarchical Bayesian Modeling

- Developed a computational pipeline integrating physiological signals (pupillometry) with Drift Diffusion Models (DDM) to decouple decision bias from information processing efficiency.
- Implemented hierarchical Bayesian estimation to quantify how physiological arousal mediates cognitive load in real-time decision-making tasks.

2025 **Surgical Cognitive Monitoring System**, *Dashboard Repo* | *Training Lab Repo* | *Case Study*, R, Shiny, XGBoost, Biosignal ML

- Built an enterprise-grade, real-time cognitive state monitoring dashboard for surgical performance, integrating six evidence-based biosignals (pupillometry, grip force, tremor, HRV, blink rate, ambient noise) with parameters validated against 20+ peer-reviewed studies.
- Engineered a causal ML pipeline (XGBoost classifier + Isolation Forest anomaly detection + Platt scaling) with 57 strictly causal features, Leave-One-Surgeon-Out (LOSO) cross-validation, and SHAP-based model explainability for clinical interpretability.
- Developed a companion interactive training lab implementing three cognitive theory paradigms (Adaptive Gain Theory, Resource Competition Model, Vigilance Decrement) for research and educational use.

2025 – 26 **XR Adaptive Modality System**, *GitHub Repo* | *Case Study*, XR Design, TypeScript, HCI

- Architected a research platform for adaptive XR interfaces (Hand vs. Gaze) implementing ISO 9241-9 standards and Fitts's Law target selection.
- Developed a privacy-first telemetry system and automated analysis pipeline (R/Quarto) to process complex cognitive load metrics.
- Accelerated the prototyping lifecycle using Cursor for AI-assisted code generation, enabling rapid translation of theory into functional TypeScript/Vite prototypes.

Conference Abstracts

2020 Sarwar, S., Shafiee, M., **Dastgheib, M.** , Hajighadimi, S., Ghafarian, H., Rokni, H., *Current Diagnostic Approaches to Patients with Hypernatremia at University Health Network Hospitals*, Critical Care Canada Forum 2019 Abstracts. Can J Anesth/J Can Anesth 67 (Suppl 1), 1–162 (2020)., https://doi.org/10.1007/s12630-019-01552-z

Conference Talks and Presentations

International Conference Presentations

2025 Selective Modulation of Auditory Perception and Metacognition by Hand Grip-Based Arousal in Older Adults, *Metacognition, Aging and Perception Symposium (MAPS 2025)* , Rovereto, Italy, Sun, A., **Dastgheib, M.**, Maniscalco, B., Peters, M., Zhang, W., Seitz, A.

2025 Functional MRI Connectome Fingerprinting in Older Adults, *IEEE International Symposium on Biomedical Imaging (ISBI 2025)*, Houston, TX, Ghaffari, A., Langley, J., **Dastgheib, M.**, Chen, X., Bennett, I., Hu, X.

2024 The Differential Effect of Physical Effort within Perceptual and Visual Memory Domains: Insights from Psychometric Function Analysis, *65th Annual Meeting of the Psychonomic Society*, New York City, NY, **Dastgheib, M.**, Bennett, I., Seitz, A.

2023 The Effect of Physical Effort on Perception and Visual Memory, *International Conference on Learning and Memory*, Huntington Beach, CA, **Dastgheib, M.**, Yaghoubi, K., Kobaissi, H., Alba, J., Bennett, I., Seitz, A.

2019 The Effects of Self-guided Meditation and Napping on Non-Declarative and Declarative Memory Consolidation, *Canadian Society for Brain, Behaviour, & Cognitive Sciences meeting*, Waterloo, ON, **Dastgheib, M.**, Legro L., Stewart M., Dringenberg H.

2019 Discrepancy between Clinical Volume Assessment vs. Laboratory Investigations and Change in the Hyponatremia Managemen, *17th International Congress of Nephrology, Dialysis, and Transplantation*, Tabriz, IRN, Shafiee, M., Mumtaz, M., **Dastgheib, M.**, Rezaei, R.

2019 Common Etiologies of Hypernatremia and Current Analytic Approaches Used at University Health Network Hospitals, *17th International Congress of Nephrology, Dialysis, and Transplantation*, Tabriz, IRN, Shafiee, M., **Dastgheib, M.**, Sarwar, S., Hajighadimi, S., Rezaei, R.

Intramural Conference Presentations

2019 Current Diagnostic Approaches to Patients with Hypernatremia at University Health Network Hospitals, *Critical Care Canada Forum (CCCF)*, Toronto, ON., Sarwar, S., Shafiee, M., **Dastgheib, M.**, Hajighadimi, S., Ghafarian, H., Rokni, H.

2018 The Effects of Self-guided Meditation and Napping on Non-Declarative and Declarative Memory Consolidation, *Southern Ontario Neuroscience Association (SONA) conference*, Guelph, ON., **Dastgheib, M.**, Legro L., Stewart M., Dringenberg H.

2017 Are Older Adults' Attitudes about Aging Influenced by a Psychoeducational Program?, *Psychology Undergraduate Thesis Poster Day at York University*, Toronto, ON., **Dastgheib, M.**

2016 Syndrome of Inappropriate Anti-Diuretic Hormone (SIADH); Risks and Dangers of a Misnomer, *2016 Critical Care Canada Forum (CCCF)*, Toronto, ON., Shafiee M., Tabataba-Vakili S., Parastandechehr G., Hakimfaal S., Umar S., **Dastgheib M.**, Nazarian, A., Ghaffarian H.

2015 Enzymatic Activity of Pancreatic α-amylase in Microgravity, *2015 Science Rendezvous at Ryerson University*, Toronto, ON., Aziz, T., Calabia, D., **Dastgheib, M.**, Dastgheib, M., Farache, R., Geramy, S., Haberfellner, C., Lee, J., Tampus, J., Upshaw, T

Teaching & Mentorship

Teaching Experience

2023–Present **Graduate Teaching Assistant**, *Department of Psychology*, UC Riverside
Taught discussion sections, Marked assignments and exams.

- Psychological Methods: Statistics (Prof. Josh Murillo, Spring 2026)
- Introduction to Psychology (Prof. Kelly Huffman, Winter 2026)
- Pandemic Biopsychology: from virus to vaccine (Prof. Kelly Huffman, Fall 2025)
- Neuroscience Of Learning & Memory, 2 sections (Prof. Edward Korzus, Summer 2025)
- Cognitive Processes (Prof. Halle Dimsdale-Zucker, Winter 2025 and Spring 2025)
- Psychological Methods: Statistics (Prof. Ilana Bennett, Fall 2024)
- Perception (Prof. Lawrence Rosenblum, Spring 2024)
- Introduction to Psychology (Prof. Annie Ditta, Winter 2024)
- Perception (Prof. Sarah Maples, Fall 2023)

2018 – 2019 **Graduate Teaching Assistant**, *Department of Psychology*, Queen's University
Taught classes and tutorials, Marked assignments and exams.

- History of Modern Psychology (Prof. Cella Olmstead, Winter 2019)
- Brain & Behaviour I (Prof. Tyson Baker, Fall 2019)
- Introduction to Comparative Cognition (Prof. Cella Olmstead, Fall 2019)
- Introduction to Cognition (Prof. Monica Castelhano, Winter 2018 and Winter 2019)
- Advanced Perception (Prof. Niko Troje, Winter 2018)

2011 – 2012 **Instructor**, *Mofid Middle School*, Tehran, IRN
Supervisor: Mostafa Moeini

- Persian Literature and Composition

Mentorship Experience

2025 **Affirmative Action Compliance Liaison**, *Department of Psychology*, UC Riverside

- Served on search committee to review, interview, and evaluate candidates for Laboratory Assistant position
- Reviewed and scored over 50 applicant files, ensuring fair assessment and compliance with university affirmative action policies
- Collaborated with faculty and HR to evaluate candidate qualifications and research experience
- Assessed applicants' skills in participant recruitment, neuroimaging data management, and human subjects research

2022–Present **Undergraduate Mentorship**, *UC Riverside*

- Guided undergraduate research projects resulting in conference presentations
- Supervised Jacqueline Alba's RISE Summer Program project on LC-NE system and perceptual/memory systems (oral presentation, August 2022)
- Participant recruitment and training
- Conducting behavioral experiments
- Data cleaning, analysis, and quality control (QC) of pupillometry and neuroimaging data
- Mentees: Jacqueline Alba, Hadi Kobaissi, Shanya Sanof, Haley Martinez, Niku Nagpal

2023–2024 **Undergraduate Mentorship**, *Northeastern University*

- Quality control of behavioral performance and pupillometry data
- Data cleaning and analysis
- Mentees: *Thomas Kato, Sadhana Gopinath, Rachel Keltz*

2017–2019 **Undergraduate Mentorship**, *Queen's University*

- Participant recruitment and training
- Polysomnography analysis
- Data cleaning, analysis, and quality control of EEG data
- Mentees:*Liza Legro, Katie McGuire*

Honors & Awards

2023, 2024 **Graduate Student Travel Award**, *Graduate Student Association*, UC Riverside

2021 **Dean's Distinguished Fellowship Award**, *Graduate Division*, UC Riverside

2018, 2019 **Graduate Student Conference Travel Award**, *Department of Psychology*, Queen's University

2018 **Mitchell and Wilda Andriesky Award**, *Faculty of Graduate Studies*, Queen's University

2017 **Entrance Scholarship**, *Faculty of Graduate Studies*, Queen's University

2016 **Dean's Honour Roll**, *Faculty of Science*, York University

2012 **Dean's Honour Roll**, *Faculty of Liberal Arts and Sciences*, Wichita State University

Professional Development

User Experience and Interaction Design

2024 **Interaction Design Specialization**, *LinkedIn Learning*

- Foundational training in user-centered design principles, including user research methodologies, prototyping, usability testing, and heuristic evaluation.

View Specialization

Data Science & Analytics

2018–2022 **Data Science & Analytics Training**, *UC Riverside & DataCamp*

- **Principles of Data Science (UC Riverside):** Formal graduate training in creating robust, reproducible, and automated data workflows using R, RMarkdown, and Git/GitHub. Coursework focused on data wrangling, literate programming, and transparent open-science practices.
- **Bayesian Modeling & Statistical Inference:** Extensive specialized training in Bayesian regression (`rstanarm`), Generalized Linear Models (GLMs), and factor analysis.
- **Advanced R for Reproducible Research:** Developed expertise in building R packages for sharable code, advanced object-oriented programming, and writing high-performance R code for efficiency.
- **Full Data Pipeline Proficiency:** Mastered end-to-end data processing, including data manipulation (Tidyverse, SQL), data visualization (`ggplot2`), and machine learning (unsupervised and supervised learning in R).

Research Methodology and Tools

2021 **Research Transparency & Reproducibility Training (RT2)**, *Berkeley Initiative for Transparency in the Social Sciences (BITSS)*

- Intensive, hands-on training in cutting-edge methods for transparent and reproducible research.
- **Key Competencies:** Pre-registration and pre-analysis plans to reduce bias; data de-identification for ethical data sharing; version control with Git/GitHub; creation of dynamic documents in R for computational reproducibility.
- View Certificate

2021 **FSL Neuroimaging Analysis Training**, *Oxford Centre for Functional MRI of the Brain (FMRIB)*

- Practical training in fMRI analysis, including preprocessing, GLM-based statistical modeling, and functional connectivity.
- View Certificate

Clinical Assessment Tools

2021 **Montreal Cognitive Assessment (MoCA) Certification**, *MoCA Cognitive*

- Certified to administer and interpret the MoCA test for cognitive impairment screening

Community Involvement

Riverside, CA

2024 **Speaker, Aging and Your Brain Symposium**, *Center for Health Disparities Research*, UC Riverside

- Delivered a presentation on complex data analysis techniques to a lay audience of 50+ community members to improve science literacy and engagement
- Conducted a Q&A session for older adults

2022 – 2023 **Lab Representative**, *Psi Chi Undergraduate Lab Fair*, UC Riverside

2022 **Panelist in the cognitive neuroscience area**, *UCR R'PSYC Conference,* , UC Riverside

2022, 2024 **Reviewer**, *UCR R'PSYC Conference*, UC Riverside

- Reviewing and assessment of abstract submissions

Kingston, ON

2018 – 2019 **Category Judge**, *The Frontenac, Lennox and Addington Science Fair*

- Evaluated 50+ projects at the science fair

2018 – 2019 **Content Developer**, *RISE — Research and Information Science Education, Queen's University*

- Improved research skills and scientific literacy in students
- Educational outreach at the secondary school level

2018 – 2019 **Volunteer**, *Providence Care Hospital*

2017 – 2019 **Educational Mentorship**, *QchaT Psychology, Queen's University*

Tehran, IRN

2008 – 2012 **Columnist**, *Music review in Aan magazine*, University of Tehran Student Journal

Affiliations & Memberships

Current	Psychonomic Society, Cognitive Science Society, The National Center for Faculty Development and Diversity
Past	Southern Ontario Neuroscience Association (SONA), Canadian Society for Brain, Behaviour, & Cognitive Science (CSBBCS), Society for Philosophy and Psychology